

SEO OS - Technology Stack & Applications Used

This document summarizes the applications, frameworks, cloud services, AI tools, and technologies used to design, develop, deploy, and run the SEO OS platform.

Design & Planning

Application	Purpose
Figma	UI/UX design, wireframes, components
Cursor AI	Primary AI-assisted coding environment
ChatGPT	Architecture, planning, prompts, debugging, product strategy

Frontend

Technology	Purpose
React	Frontend framework
TypeScript	Programming language
Vite	Build tool
Tailwind CSS	Styling
shadcn/ui	UI components
Framer Motion	Animations
React Router	Routing

Backend

Technology	Purpose
Node.js	Runtime
Express.js	REST API
TypeScript	Backend language
pg-boss	Background jobs & queues

Database & Auth

Technology	Purpose
Supabase	Backend platform

PostgreSQL	Database
Supabase Auth	Authentication
Supabase Storage	File storage
Row Level Security	Multi-tenant security

Artificial Intelligence

Technology	Purpose
Google Gemini	Primary AI model
Ollama	Local AI fallback
Custom AI Runtime	Agent orchestration
Knowledge Engine	RAG
AI Memory	Persistent context

Deployment & DevOps

Platform	Purpose
GitHub	Source control
Railway	Backend hosting
Netlify	Frontend hosting
Docker	Containerization
TurboRepo	Monorepo
npm Workspaces	Package management
GitHub Actions	CI/CD foundation

Major SEO OS Modules

Module
Mission Control
Knowledge Engine
Browser Intelligence
SEO Intelligence
Campaign Engine
Backlink Builder

Backlink Automation
Relationship Intelligence
Outreach Engine
Executive Dashboard
Demo Mode

Skills Gained

Area
AI Agent Architecture
SaaS Development
React & TypeScript
Supabase & PostgreSQL
REST APIs
Authentication & Authorization
Cloud Deployment
Git & DevOps
Enterprise Dashboard Design
Product Architecture

Deployment Architecture

GitHub → Railway (Backend API) + Netlify (Frontend) → Supabase (Database, Authentication, Storage) → Google Gemini (AI) with Ollama as an optional fallback. The application is organized as a TurboRepo monorepo with shared packages for AI runtime, providers, SEO intelligence, campaign engine, and backlink management.